



Amrita School of AI
COIMBATORE CAMPUS

UNIT 3 · SESSIONS 14 AND 15

Many qubits and observables

Two qubits, entanglement, and what a “measurable quantity” is. Then the whole course on four slides.

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SESSION 14

Two qubits

Postulate 4: joint systems live in a tensor product.

Two qubits live in $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$.

Four amplitudes, one for each pair of outcomes.

NAMING THE BASIS

$|ab\rangle = |a\rangle \otimes |b\rangle$, **first symbol is the first qubit**

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Entries 1 to 4 are always in this order: 00, 01, 10, 11. It is counting in binary.

Building basis states and a product state

$$|10\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 \\ 0 \cdot 0 \\ 1 \cdot 1 \\ 1 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

A state of the form $|a\rangle \otimes |b\rangle$ is a **product state**.

A product state, measured

$$|+\rangle \otimes |-\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

$$p(ab) = |\langle ab|\psi\rangle|^2 \quad \implies \quad p(00) = p(01) = p(10) = p(11) = \frac{1}{4}$$

Measuring both qubits: the Born rule, with four outcomes instead of two.

$X \otimes I$: flip the first qubit, leave the second alone

$$X \otimes I = \begin{pmatrix} 0 \cdot I & 1 \cdot I \\ 1 \cdot I & 0 \cdot I \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$(X \otimes I)|01\rangle = (X|0\rangle) \otimes (I|1\rangle) = |1\rangle \otimes |1\rangle = |11\rangle$$

Which qubit does the gate act on?

- **Two common slips**

Writing $I \otimes X$ for “flip the first qubit”. That flips the **second**.

Multiplying the 2×2 matrix X into a 4-entry vector. The sizes do not match.

- **The rule**

The slot in $A \otimes B$ says which qubit: A acts on the first, B on the second.

A gate on two qubits is always 4×4 .

CNOT: flip the target when the control is 1

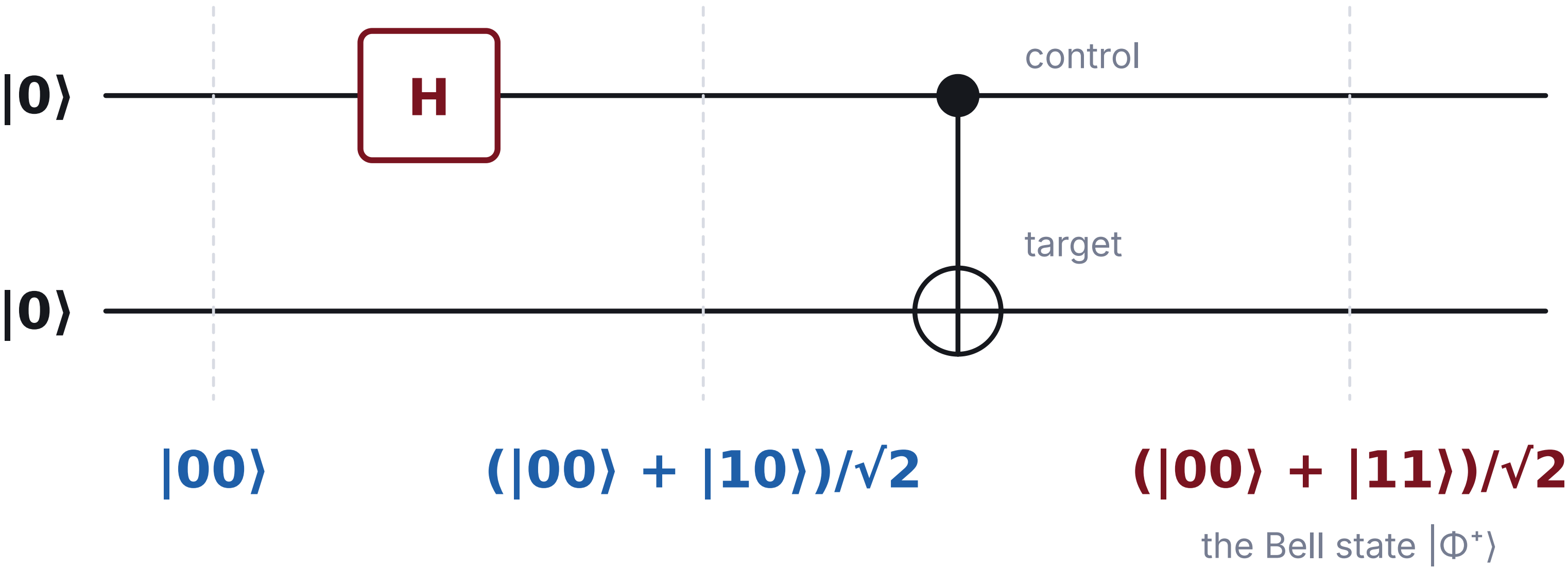
$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

IN	OUT
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

First qubit is the control, second the target.

THE MOST FAMOUS CIRCUIT

H on the first qubit, then CNOT



H on the first qubit

$$(H \otimes I)|00\rangle = (H|0\rangle) \otimes |0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Still a product state. Nothing unusual yet.

Then CNOT

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = |\Phi^+\rangle$$

The $|10\rangle$ part had control 1, so its target flipped: $|10\rangle \rightarrow |11\rangle$. This is the **Bell state** $|\Phi^+\rangle$.

A two-qubit state that is not a product is entangled.

It cannot be written as $|a\rangle \otimes |b\rangle$ for any single-qubit $|a\rangle, |b\rangle$.

$|\Phi^+\rangle$ is not a product

$$(a_0|0\rangle + a_1|1\rangle) \otimes (b_0|0\rangle + b_1|1\rangle) = a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle$$

$$\text{Need: } a_0b_0 = a_1b_1 = \frac{1}{\sqrt{2}}, \quad a_0b_1 = 0, \quad a_1b_0 = 0$$

From $a_0b_1 = 0$, either $a_0 = 0$ (then $a_0b_0 = 0$) or $b_1 = 0$ (then $a_1b_1 = 0$). Both contradict the first line.

Product exactly when $c_{00}c_{11} = c_{01}c_{10}$

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle$$

$$|\Phi^+\rangle : c_{00}c_{11} = \frac{1}{2} \neq 0 = c_{01}c_{10} \implies \text{entangled}$$

$$|+\rangle \otimes |-\rangle : c_{00}c_{11} = -\frac{1}{4} = c_{01}c_{10} \implies \text{product}$$

Measure both qubits of $|\Phi^+\rangle$

OUTCOME	PROBABILITY
00	$\frac{1}{2}$
01	0
10	0
11	$\frac{1}{2}$

Each qubit on its own is a fair coin.

Yet the two results **always agree**. After the first qubit reads 0, the state is $|00\rangle$, and the second reads 0 with certainty.

MYTH

"Entanglement sends signals faster than light"

- **The myth**

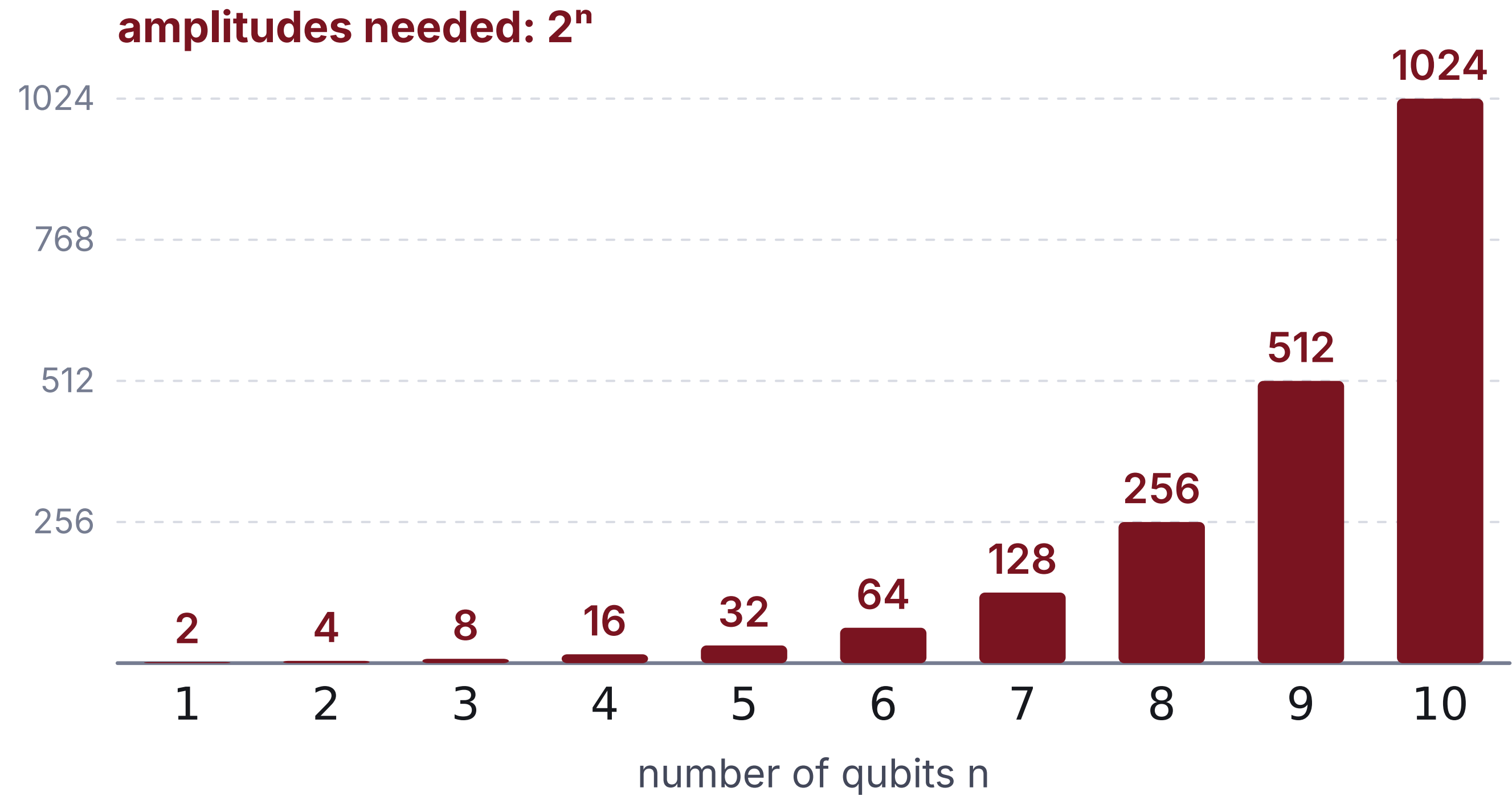
Measuring my qubit instantly sets yours, so I can send you a message.

- **What the mathematics says**

Your results alone are a fair coin, whatever I do to my qubit.

The agreement shows up only when we compare lists, and comparing needs an ordinary message.

n qubits need 2^n amplitudes



At 16 bytes per complex amplitude

16 KB

10 qubits: 1 024 amplitudes

17 GB

30 qubits: about a billion

18 PB

50 qubits: about 10^{15}

Your Part C simulator stores the full vector. A typical laptop runs out of memory before 30 qubits.

TRY THIS NOW · 5 MINUTES

Entangled or not?

1. Apply CNOT to $|+\rangle \otimes |1\rangle$.
2. Apply CNOT to $|1\rangle \otimes |+\rangle$.
3. Which result is entangled? Use the shortcut to check.

Check yours

$$\text{CNOT} \frac{1}{\sqrt{2}} (|01\rangle + |11\rangle) = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \quad c_{00}c_{11} = 0 \neq \frac{1}{2} = c_{01}c_{10} : \text{ entangled}$$

$$\text{CNOT} \frac{1}{\sqrt{2}} (|10\rangle + |11\rangle) = \frac{1}{\sqrt{2}} (|11\rangle + |10\rangle) = |1\rangle \otimes |+\rangle : \text{ product}$$

CNOT does not always entangle. Here the target $|+\rangle$ is unchanged by a flip.

Two qubits in five lines

```
import numpy as np
H = np.array([[1, 1], [1, -1]]) / np.sqrt(2)
I = np.eye(2)
CNOT = np.array([[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 0, 1], [0, 0, 1, 0]])
ket00 = np.kron([1, 0], [1, 0])
print(CNOT @ np.kron(H, I) @ ket00)    # [0.707 0 0 0.707]
```

`np.kron` is the tensor product. Order matters, exactly as on paper.

What to do before next session

- Redo worked example 3 without looking.
- Assignment Part C: add `np.kron`, CNOT, and a function that measures two qubits.
- Revise eigenvalues from session 9. Session 15 is built on them.

SESSION 15

Observables

Postulate 3 again, written in the language of Hermitian matrices.

A measurable quantity is a Hermitian matrix.

Physicists call it an **observable**. Energy, spin and position are all observables.

From a Hermitian matrix to a measurement

$$A|a_k\rangle = \lambda_k|a_k\rangle, \quad p(\lambda_k) = |\langle a_k|\psi\rangle|^2$$

1. The possible outcomes are the eigenvalues λ_k : real, because A is Hermitian.
2. The eigenvectors $|a_k\rangle$ form an orthonormal basis: measure in it.
3. Afterwards the state is $|a_k\rangle$.

Z as an observable

$$Z|0\rangle = +1|0\rangle, \quad Z|1\rangle = -1|1\rangle$$

Eigenvalues $+1$ and -1 , eigenvectors $|0\rangle$ and $|1\rangle$. Measuring Z is measuring in the basis $|0\rangle, |1\rangle$, with the outcomes labelled $+1$ and -1 in place of 0 and 1.

X as an observable

$$X|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle, \quad X|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -|-\rangle$$

Eigenvalues ± 1 again, eigenvectors $|\pm\rangle$. Measuring X is measuring in the $\pm$ basis.

Every observable splits into its measurement

$$A = \sum_k \lambda_k |a_k\rangle\langle a_k|$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad X = |+\rangle\langle +| - |-\rangle\langle -|$$

Check the second: $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$

The average over many measurements

$$\langle A \rangle = \sum_k \lambda_k p(\lambda_k) = \langle \psi | A | \psi \rangle$$

The first form needs the eigenvectors. The second, the “sandwich”, needs only one matrix product and one inner product.

WORKED EXAMPLE 6 · TWO WAYS

$$\langle Z \rangle \text{ for } |\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$$

$$\text{Average: } (+1) \cdot \frac{3}{4} + (-1) \cdot \frac{1}{4} = \frac{1}{2}$$

$$\text{Sandwich: } Z|\psi\rangle = \frac{1}{2} \begin{pmatrix} \sqrt{3} \\ -1 \end{pmatrix}, \quad \langle\psi|Z|\psi\rangle = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \left(-\frac{1}{2}\right) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

WORKED EXAMPLE 7 · SAME STATE

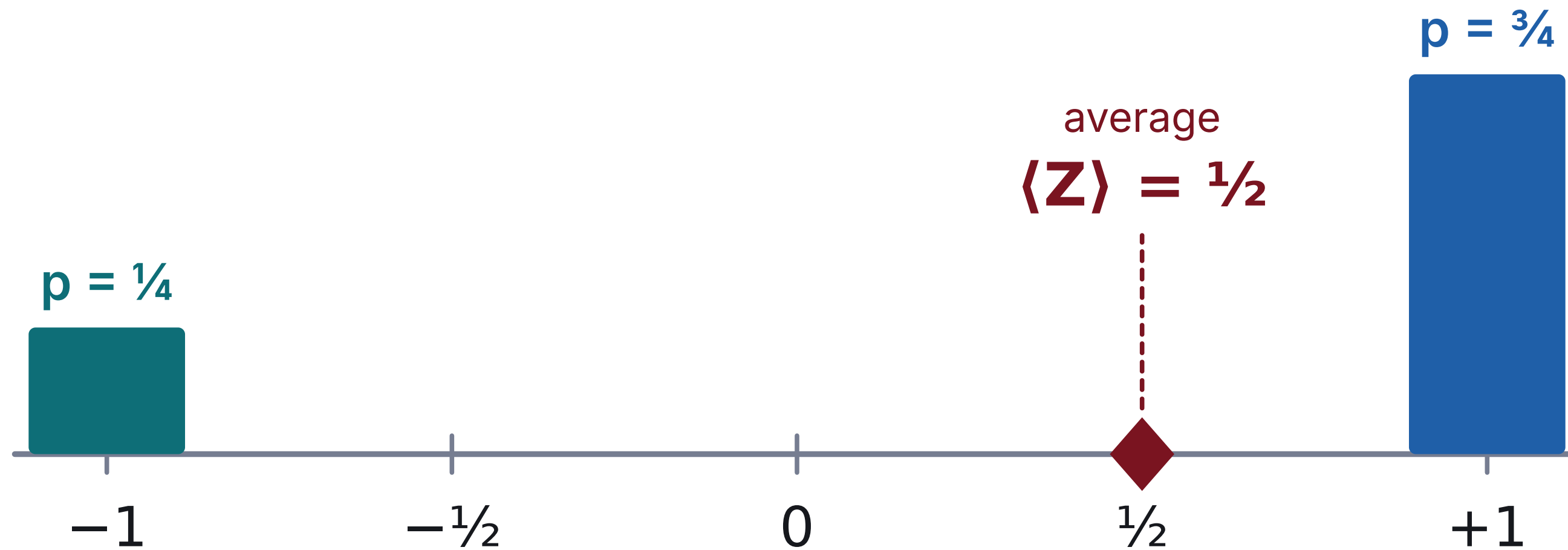
$$\langle X \rangle \text{ for } |\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$$

$$X|\psi\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}, \quad \langle\psi|X|\psi\rangle = \frac{\sqrt{3}}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2} \approx 0.866$$

$$\text{Check with session 12: } p(+)-p(-) = \frac{2+\sqrt{3}}{4} - \frac{2-\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

PITFALL

The expectation value is an average, not an outcome



Each single measurement gives -1 or $+1$. Never $1/2$.

TRY THIS NOW · 5 MINUTES

Three expectation values for $|+i\rangle$

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Compute $\langle Z \rangle$, $\langle X \rangle$ and $\langle Y \rangle$ with the sandwich. Remember to conjugate the bra.

ANSWERS

$$\langle Z \rangle = 0, \langle X \rangle = 0, \langle Y \rangle = 1$$

$$Z|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} : \frac{1}{2} (1 + (-i)(-i)) = 0 \quad X|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix} : \frac{1}{2} (i + (-i)) = 0$$

$$Y|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} (-i)(i) \\ i \end{pmatrix} = |\psi\rangle \implies \langle Y \rangle = \langle \psi | \psi \rangle = 1$$

$|+i\rangle$ is an eigenvector of Y with eigenvalue $+1$.

COURSE REVIEW

The whole course on four postulates

Every tool from Units 1 and 2 appears in at least one of them.

State and evolution

1. State

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \langle\psi|\psi\rangle = 1$$

A unit ket. The kets $e^{i\gamma}|\psi\rangle$ and $|\psi\rangle$ are one state.

2. Evolution

$$|\psi\rangle \rightarrow U|\psi\rangle, \quad U^\dagger U = I$$

Gates are unitary. A sequence A then B is BA .

Measurement and composite systems

3. Measurement

$$p(k) = |\langle b_k | \psi \rangle|^2, \quad \langle A \rangle = \langle \psi | A | \psi \rangle$$

Afterwards the state is $|b_k\rangle$. Observables are Hermitian, outcomes are eigenvalues.

4. Composite systems

$$|ab\rangle = |a\rangle \otimes |b\rangle, \quad \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$$

States that are not products are entangled.

WHERE EACH TOOL WENT

Units 1 and 2, put to work

FROM	TOOL	BECAME
Unit 1	complex numbers, modulus	amplitudes and $ \alpha ^2$
Unit 1	Euler's formula, phase	global and relative phase
Unit 2	inner product	the Born rule
Unit 2	unitary matrices	gates
Unit 2	Hermitian matrices, eigenvalues	observables and outcomes
Unit 2	tensor product	many qubits, entanglement

The expectation value is a sandwich

```
import numpy as np
X = np.array([[0, 1], [1, 0]])
Y = np.array([[0, -1j], [1j, 0]])
Z = np.array([[1, 0], [0, -1]])
psi = np.array([1, 1j]) / np.sqrt(2)      # |+i>
for A in (Z, X, Y):
    print(np.vdot(psi, A @ psi).real)      # 0.0, 0.0, 1.0
print(np.linalg.eigvalsh(Y))              # [-1.  1.]
```

Finishing the course

- **Assignment Part C**, the qubit simulator in NumPy, is due at the end of this week.
- **Viva** on your workbook, in this week or the next.
- **End-semester exam**: the whole syllabus, all three units. Worked examples from every deck are fair game.